

Conjecture 00000004285 is false: a countable 2-free, not 3-free abelian group

Jackmeson1

1 The conjecture and the reading

Statement (English text of the repository). “Definition: n -free means every subset of size at most n is contained in a free pure subgroup. Conjecture: The smallest cardinality of a group that is n -free but not $n+1$ -free is aleph with subscript $n-1$, the cardinal spectrum is strictly increasing, and the bound is optimal.” The Chinese text says the same.

Reading. The text defines n -free with “subset of size at most n ”, so n is a natural number and the subsets are finite. Groups are abelian, since purity is a notion of abelian group theory. “Free” means free abelian, i.e. a free $\mathbb{Z}$ -module (Wikipedia, *Free abelian group*: “a free abelian group is an abelian group with a basis”). “Pure” is the standard notion (Wikipedia, *Pure subgroup*: “In mathematics, especially in the area of algebra studying the theory of abelian groups, a pure subgroup is a generalization of direct summand”):

Definition 1. A subgroup $H \leq A$ is *pure* if $mA \cap H \subseteq mH$ for every $m \in \mathbb{Z}$ (the reverse inclusion always holds). A is *n -free* if every finite $S \subseteq A$ with $|S| \leq n$ lies in a pure subgroup H that is a free abelian group.

The conjecture is a conjunction. Its first part (**MinCardClaim**) says: for every $n \geq 1$, $\aleph_{n-1}$ is the least cardinality of an abelian group that is n -free but not $(n+1)$ -free. We refute this part at $n = 2$, so the conjunction is false. We work with groups in Lean’s universe **Type**. Our counterexample lies there, so the refutation also applies to any universe-lifted version of the claim.

2 The counterexample

Let p_k be the k^3 -th prime, counting from 0 (Lean: `Nat.nth Nat.Prime (k^3)`). The p_k are distinct primes with $p_k \geq k^3$. Put $w_k = (1, k, k^2)/p_k \in \mathbb{Q}^3$ and

$$G = \mathbb{Z}^3 + \sum_{k \geq 0} \mathbb{Z} w_k \subseteq \mathbb{Q}^3.$$

G is countable. For $a \in \mathbb{Z}^3$ write $q_a(k) = a_0 + a_1 k + a_2 k^2$.

Lemma 2 (`bad_finite`). *If $a \neq 0$, then $\{k : p_k \mid q_a(k)\}$ is finite.*

Proof. Let $A = |a_0| + |a_1| + |a_2|$. If $k \geq A + 2$, $p_k \mid q_a(k)$ and $q_a(k) \neq 0$, then $k^3 \leq p_k \leq |q_a(k)| \leq A k^2 < k^3$, which is impossible. The nonzero polynomial q_a has finitely many roots. $\square$

Theorem 3 (`G_not_three_free`). *G is not 3-free.*

Proof. Let H be a pure subgroup containing e_0, e_1, e_2 . Since $p_k w_k = e_0 + k e_1 + k^2 e_2 \in H$, purity and torsion-freeness give $w_k \in H$ for all k . Suppose H were free. Take a basis b of H . Since $e_0 \neq 0$, some coordinate functional $f = b_i^*: H \rightarrow \mathbb{Z}$ has $f(e_0) \neq 0$. With $a_j = f(e_j)$ we get $q_a(k) = f(p_k w_k) = p_k f(w_k)$, so $p_k \mid q_a(k)$ for every k . By the lemma, $a = 0$. This contradicts $f(e_0) \neq 0$. $\square$

Theorem 4 (`G_two_free`). *G is 2-free.*

Proof. Let $S \subseteq G$ with $|S| \leq 2$. The $\mathbb{Q}$ -linear map $\mathbb{Q}^3 \rightarrow \mathbb{Q}^S$, $x \mapsto (x \cdot s)_s$, has a nonzero kernel. Clearing denominators gives $N \in \mathbb{Z}^3 \setminus 0$ with $N \cdot s = 0$ for $s \in S$ (`normal_vector`). Let $H = \{g \in G : N \cdot g = 0\}$. Then $S \subseteq H$. H is pure: if $m \neq 0$ and $N \cdot (mg) = 0$, then $g \in H$ (`pure_ker`).

Bounded denominators (`key_dvd`, `bounded_den`). Let B be the finite set of k with $p_k \mid q_N(k)$, and put $M = \prod_{k \in B} p_k$. Write $g = z + \sum_{k \in F} c_k w_k$ with $z \in \mathbb{Z}^3$ and $N \cdot g = 0$, that is, $N \cdot z + \sum_{k \in F} c_k q_N(k) = 0$. Multiplying by $\prod_{k \in F} p_k$ and reducing modulo p_j shows that $p_j \mid c_j q_N(j) \prod_{k \in F \setminus j} p_k$. The primes are distinct, so for $j \notin B$ we get $p_j \mid c_j$. Hence $Mg \in \mathbb{Z}^3$.

So $g \mapsto Mg$ embeds H into $\mathbb{Z}^3$. Therefore H is finitely generated (it is a submodule of a Noetherian module). It is torsion-free, so it is free (`Mathlib.Module.free_of_finite_type_torsion_free`) (`free_ker`). $\square$

Theorem 5 (`not_minCardClaim`). *MinCardClaim is false: G is a 2-free, not 3-free group with $|G| \leq \aleph_0 < \aleph_1 = \aleph_{2-1}$.*

Remarks. (1) For $n = 1$ the claimed value $\aleph_0$ is not contradicted. Our result concerns $n = 2$. (2) Readings that are *not* refuted here: (a) a cardinal-indexed reading (“ κ -free”: subsets of size $< \aleph_n$). Under it, a group of size $\aleph_{n-1}$ that is $\aleph_n$ -free lies in a free pure subgroup, so it is free and hence $(n+1)$ -free; the claim would fail, but this is not formalized. (b) Readings with non-abelian groups or “free” meaning free (non-abelian) group.

3 Lean formalization

File `Conjecture4285/Basic.lean` (namespace `C4285`, about 400 lines, only `import Mathlib`). Definitions: `IsPureSubgroup`, `NFree`, `p`, `q`, `w`, `G`, `MinCardClaim`. Main results:

- `G_two_free : NFree 2 GT`, `G_not_three_free : ~NFree 3 GT`;
- `exists_countable_two_free_not_three_free`: there is $A : \text{Type}$ with an `AddCommGroup` structure, $\#A \leq \aleph_0$, `NFree 2 A` and `~NFree 3 A`;
- `not_minCardClaim : ~MinCardClaim`, where `MinCardClaim` is `forall n, 1 <= n -> IsLeast {k | exists (A : Type) (_ : AddCommGroup A), #A = k /\ NFree n A /\ ~NFree (n+1) A} (aleph ((n-1 : Nat) : Ordinal))`.

Axiom audit. `#print axioms for C4285.not_minCardClaim and C4285.exists_countable_two_free_not_three_free` gives `[propext, Classical.choice, Quot.sound]`. There is no `sorry` and no `native_decide`. Lean toolchain v4.33.1, Mathlib v4.33.1.